



ASSESSMENT PLAN

FOR

Data Mining and Machine Learning Fundamentals for Beginners

Course Code: TGS-2020503264

Conducted by Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 17.0 · 10 September 2026

Document Version Control Record

Version	Effective date	Summary of changes	Author
17.0	10 September 2026	Assessment plan aligned to a 10-question WA (K1–K10) and seven-task Python/Streamlit PP (A1–A9), replacing legacy tool workflows.	Dr Alfred Ang

Assessment Strategy

Written Assessment	10 open-ended short-answer questions · K1–K10 · 60 minutes · open book
Practical Performance	7 connected tasks · A1–A9 · 60 minutes · open book
Allowed references	Course slides, Learner Guide, lab guides, and supplied mock data
Submission	Upload completed evidence to https://lms-tms.tertiaryinfotech.com/
Competency decision	All prescribed assessment requirements must be met; assessor records C or NYC

Written Assessment Mapping

Question	Knowledge	Course evidence	Expected response evidence
1	K1	Topic 2	Missingness treatment and validation
2	K2	Topic 3	Ridge/Lasso and overfitting
3	K3	Topic 5	Dendrogram cut and silhouette
4	K4	Topic 1	Supervised vs unsupervised boundary
5	K5	Topics 3–5	Model selection and visual evidence
6	K6	Topic 4	Precision, recall and F1
7	K7	Topic 4	ROC/AUC and threshold policy
8	K8	Topic 6	PCA retained variance
9	K9	Topic 7	Support, confidence and lift
10	K10	Topics 2–3	Scaling and statistical assumptions

Practical Performance Mapping

Task	Abilities	Aligned labs	Required artifact
1	A1	Labs 1–3	Audited, cleaned, joined dataset
2	A2	Labs 4–6	Regression metrics and actual/predicted evidence
3	A3/A4	Lab 14	Frequent itemsets and interpretable rules
4	A5	Labs 10–12	PCA plus clustering output
5	A6	Lab 11	Dendrogram cut and coloured scatter
6	A7	Labs 7–8	Three-classifier experiment
7	A8/A9	Lab 9	Metric comparison,

			threshold decision, and visual explanation
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